

B2: Symmetry and Relativity — Model Solutions, Trinity Term 2022

Signature $(-, +, +, +)$; $x^\mu = (ct, x, y, z)$, $x_\mu = (-ct, x, y, z)$; $\partial_\mu \equiv \partial/\partial x^\mu$, $\partial^\mu \equiv \partial/\partial x_\mu$.

Question 1 — Lorentz covariance of Maxwell's equations

Problem. Identify ∂^μ vs ∂_μ among the operators in (3); show $\partial_\mu \partial^\mu$ is invariant; derive charge conservation and the 4-current; in Lorenz gauge show $\partial^\nu \partial_\nu A^a = -\mu_0 J^a$ and prove A^μ is a 4-vector; for a plane wave show one can choose $\vec{k} \cdot \vec{\epsilon} = 0$.

(a)

Apply the chain rule to a scalar Φ :

$$\frac{\partial \Phi}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu} \frac{\partial \Phi}{\partial x^\nu}.$$

Read off the transformation of $\partial/\partial x^\mu$:

$$\partial'_\mu = \frac{\partial x^\nu}{\partial x'^\mu} \partial_\nu.$$

Compare with covariant rule (2): $\partial/\partial x^\mu$ transforms like x_μ . Hence

$$\partial_\mu = \left(\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

Raise with $\eta^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$:

$$\partial^\mu = \left(-\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

(b)

Apply the chain rule twice:

$$\partial'_\mu \partial'^\mu = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\mu}{\partial x^\beta} \partial_\alpha \partial^\beta.$$

The bracketed product is the inverse-matrix identity:

$$\frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x'^\mu}{\partial x^\beta} = \delta^\alpha_\beta.$$

Contract:

$$\partial'_\mu \partial'^\mu = \partial_\alpha \partial^\alpha.$$

(c)

Take divergence of Ampère–Maxwell, use $\nabla \cdot (\nabla \times \vec{B}) = 0$:

$$0 = \mu_0 \nabla \cdot \vec{J} + \frac{1}{c^2} \frac{\partial}{\partial t} (\nabla \cdot \vec{E}).$$

Use Gauss $\nabla \cdot \vec{E} = \rho/\epsilon_0$ and $c^2 = 1/(\mu_0 \epsilon_0)$:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0.$$

Match to $\partial_\mu J^\mu = 0$ with $\partial_\mu = ((1/c)\partial_t, \nabla)$:

$$\boxed{J^\mu = (c\rho, \vec{J}), \quad \partial_\mu J^\mu = 0.}$$

∂_μ is covariant by (a); quotient rule then makes J^μ a 4-vector.

(d)

Insert $\vec{E} = -\nabla\phi - \partial_t \vec{A}$ into Gauss:

$$-\nabla^2 \phi - \frac{\partial}{\partial t} (\nabla \cdot \vec{A}) = \rho/\epsilon_0.$$

Insert potentials into Ampère–Maxwell; use $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$:

$$-\nabla^2 \vec{A} + \nabla(\nabla \cdot \vec{A}) + \frac{1}{c^2} \frac{\partial}{\partial t} \left(\nabla\phi + \frac{\partial \vec{A}}{\partial t} \right) = \mu_0 \vec{J}.$$

Lorenz condition $\partial_\mu A^\mu = 0$ with $A^\mu = (\phi/c, \vec{A})$:

$$\frac{1}{c^2} \frac{\partial \phi}{\partial t} + \nabla \cdot \vec{A} = 0.$$

Substitute into the Gauss equation:

$$\left(-\nabla^2 + \frac{1}{c^2} \partial_t^2 \right) \phi = \rho/\epsilon_0.$$

Substitute into the Ampère equation:

$$\left(-\nabla^2 + \frac{1}{c^2} \partial_t^2 \right) \vec{A} = \mu_0 \vec{J}.$$

Recognise $\partial^\nu \partial_\nu = -(1/c^2) \partial_t^2 + \nabla^2$; combine with $A^0 = \phi/c$ and $J^0 = c\rho$:

$$\boxed{\partial^\nu \partial_\nu A^a = -\mu_0 J^a.}$$

RHS is a 4-vector by (c); $\partial^\nu \partial_\nu$ is a scalar by (b); quotient rule: A^a is a 4-vector.

(e)

Substitute the plane wave into Lorenz gauge:

$$\partial_b A^b = \epsilon^b k_b \cos(k \cdot x) = 0 \implies \epsilon^b k_b = 0.$$

Residual gauge $A^\mu \rightarrow A^\mu + \partial^\mu \chi$ with $\square \chi = 0$. Choose $\chi = \chi_0 \cos(k \cdot x)$; vacuum gives $k^\mu k_\mu = 0$, so $\square \chi = 0$. Polarisation shifts:

$$\epsilon^b \longrightarrow \epsilon^b - \chi_0 k^b.$$

Choose $\chi_0 = \epsilon^0/k^0$ to set $\epsilon^0 = 0$. Lorenz then gives $\epsilon^b k_b = \vec{\epsilon} \cdot \vec{k} = 0$:

$$\boxed{\vec{k} \cdot \vec{\epsilon} = 0.}$$

Question 2 — Pion decay kinematics

Problem. ($c = 1$.) Find the CM-frame velocity for N particles; show $\sin^2(A/2) = M_\pi^2/(4E_1E_2)$ for $\pi^0 \rightarrow \gamma\gamma$; with isotropic CM emission derive dN/dA ; find the minimum and peak of the distribution.

(a)

Total 4-momentum: $P^\mu = (\sum E_i, \sum \vec{p}_i)$. CM frame defined by $\sum \vec{p}'_i = 0$. Boost by $\vec{\beta}$ along $\vec{P}_{\text{tot}}$:

$$\vec{P}'_{\text{tot}} = \gamma(\vec{P}_{\text{tot}} - \vec{\beta}E_{\text{tot}}).$$

Set $\vec{P}'_{\text{tot}} = 0$ and adopt the question's passive-boost sign:

$$\boxed{\vec{\beta} = -\frac{\sum_i \vec{p}_i}{\sum_i E_i}.$$

(b)

Conservation $p_\pi = p_1 + p_2$; square:

$$-M_\pi^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2.$$

Photons massless ($p_i^2 = 0$):

$$M_\pi^2 = -2p_1 \cdot p_2 = 2(E_1E_2 - \vec{p}_1 \cdot \vec{p}_2).$$

Use $|\vec{p}_i| = E_i$ and $\vec{p}_1 \cdot \vec{p}_2 = E_1E_2 \cos A$:

$$M_\pi^2 = 2E_1E_2(1 - \cos A).$$

Apply $1 - \cos A = 2\sin^2(A/2)$:

$$\boxed{\sin^2(A/2) = \frac{M_\pi^2}{4E_1E_2}. \quad (1)}$$

(c)

CM frame: photons back-to-back with $E_1^* = E_2^* = M_\pi/2$. Boost photon 1 (angle θ_*) and photon 2 (angle $\pi - \theta_*$):

$$E_1 = \gamma \frac{M_\pi}{2} (1 + \beta \cos \theta_*).$$

$$E_2 = \gamma \frac{M_\pi}{2} (1 - \beta \cos \theta_*).$$

Multiply:

$$E_1E_2 = \gamma^2 \frac{M_\pi^2}{4} (1 - \beta^2 \cos^2 \theta_*).$$

Insert into (1):

$$\sin^2(A/2) = \frac{1}{\gamma^2(1 - \beta^2 \cos^2 \theta_*)}. \quad (2)$$

Invert for $\cos^2 \theta_*$ (use $\gamma^2 \beta^2 = \gamma^2 - 1$):

$$\cos^2 \theta_* = \frac{\gamma^2 \sin^2(A/2) - 1}{(\gamma^2 - 1) \sin^2(A/2)}. \quad (3)$$

Differentiate (2):

$$\sin(A/2) \cos(A/2) dA = \frac{2\beta^2 \sin \theta_* \cos \theta_*}{\gamma^2(1 - \beta^2 \cos^2 \theta_*)^2} d\theta_*.$$

Use $\gamma^2(1 - \beta^2 \cos^2 \theta_*) = 1/\sin^2(A/2)$:

$$\sin(A/2) \cos(A/2) dA = 2\beta^2 \gamma^2 \sin \theta_* \cos \theta_* \sin^4(A/2) d\theta_*.$$

Solve for $d\theta_*/dA$:

$$\frac{d\theta_*}{dA} = \frac{\cos(A/2)}{2\beta^2 \gamma^2 \sin^3(A/2) \sin \theta_* \cos \theta_*}.$$

Apply CM rule $dN/d\theta_* = a \sin \theta_*$:

$$\frac{dN}{dA} = \frac{a \cos(A/2)}{2\beta^2 \gamma^2 \sin^3(A/2) \cos \theta_*}.$$

Substitute (3) and use $\beta^2 \gamma^2 = \gamma^2 - 1$:

$$\frac{1}{\beta^2 \gamma^2 \cos \theta_*} = \frac{\sin(A/2)}{\sqrt{\gamma^2 - 1} \sqrt{\gamma^2 \sin^2(A/2) - 1}}.$$

Combine; replace $\sqrt{\gamma^2 - 1} = \gamma\beta$:

$$\boxed{\frac{dN}{dA} = \frac{a}{2\gamma} \frac{\cos(A/2)}{\beta \sin^2(A/2) \sqrt{\gamma^2 \sin^2(A/2) - 1}}, \quad f(\gamma) = \frac{1}{2\gamma}.}$$

(d)

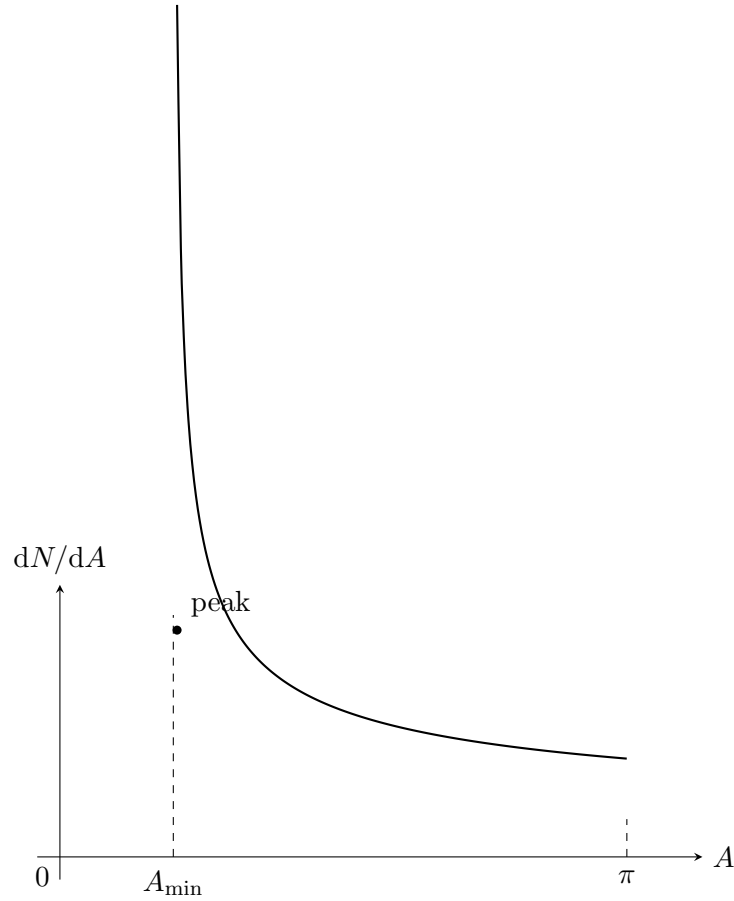
From (2), $\sin^2(A/2)$ minimised when $1 - \beta^2 \cos^2 \theta_*$ maximised, i.e. at $\cos^2 \theta_* = 0$:

$$\sin^2(A_{\min}/2) = 1/\gamma^2.$$

$$\boxed{A_{\min} = 2 \arcsin(1/\gamma).}$$

The factor $1/\sqrt{\gamma^2 \sin^2(A/2) - 1}$ diverges as $A \rightarrow A_{\min}^+$; the other factors are smooth and finite there. Hence

$$\boxed{A_{\text{peak}} = A_{\min} = 2 \arcsin(1/\gamma).}$$



Question 3 — 4-velocity, addition, two-star problem

Problem. (a) Show $d\tau = dt/\gamma_u$ and $u^\alpha = (c\gamma_u, \gamma_u \vec{u})$. (b) Find $\vec{w}$ in standard-configuration B. (c) For arbitrary boost $\vec{v}$, derive $|\vec{w}|$. (d) Stars S_1, S_2 at $c/2$ in A; light from S_2 at ω_0 received by S_1 at ω_0 : find the propagation angle in S_1 's frame, and the propagation direction in A.

(a)

Invariant interval; in rest frame $d\vec{x} = 0$:

$$-c^2 d\tau^2 = -c^2 dt^2 + d\vec{x} \cdot d\vec{x}.$$

Use $d\vec{x} = \vec{u} dt$:

$$d\tau^2 = dt^2(1 - u^2/c^2).$$

Take positive root:

$$\boxed{d\tau = dt/\gamma_u.}$$

4-velocity:

$$u^\alpha = \frac{dx^\alpha}{d\tau} = \gamma_u \frac{dx^\alpha}{dt} = \gamma_u(c, \vec{u}).$$

$$\boxed{u^\alpha = (c\gamma_u, \gamma_u \vec{u}).}$$

(b)

Boost u^α along $+x$:

$$u'^0 = \gamma_v(u^0 - \beta_v u^1), \quad u'^1 = \gamma_v(u^1 - \beta_v u^0), \quad u'^2 = u^2, \quad u'^3 = u^3.$$

Read off $\vec{w} = c\vec{u}'/u'^0$:

$$\boxed{w_x = \frac{u_x - v}{1 - vu_x/c^2}, \quad w_y = \frac{u_y}{\gamma_v(1 - vu_x/c^2)}, \quad w_z = \frac{u_z}{\gamma_v(1 - vu_x/c^2)}}.$$

(c)

Decompose along $\hat{v}$: $\vec{u}_\parallel = (\vec{u} \cdot \hat{v})\hat{v}$, $\vec{u}_\perp = \vec{u} - \vec{u}_\parallel$. Generalise (b):

$$\vec{w} = \frac{1}{1 - \vec{v} \cdot \vec{u}/c^2} \left[(\vec{u}_\parallel - \vec{v}) + \vec{u}_\perp/\gamma_v \right].$$

Square (orthogonal pieces):

$$|\vec{w}|^2 = \frac{|\vec{u}_\parallel - \vec{v}|^2 + |\vec{u}_\perp|^2/\gamma_v^2}{(1 - \vec{v} \cdot \vec{u}/c^2)^2}.$$

Insert $1/\gamma_v^2 = 1 - v^2/c^2$:

$$|\vec{u}_\parallel - \vec{v}|^2 + |\vec{u}_\perp|^2/\gamma_v^2 = |\vec{u}_\parallel - \vec{v}|^2 + |\vec{u}_\perp|^2 - (v^2/c^2)|\vec{u}_\perp|^2.$$

Combine first two via orthogonality:

$$|\vec{u}_\parallel - \vec{v}|^2 + |\vec{u}_\perp|^2 = |\vec{u} - \vec{v}|^2 = (\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}).$$

Apply hint with $\vec{u}_\parallel \times \vec{v} = 0$, so $|\vec{u} \times \vec{v}| = |\vec{u}_\perp||\vec{v}|$:

$$v^2|\vec{u}_\perp|^2 = |\vec{u} \times \vec{v}|^2 = (\vec{v} \times \vec{u}) \cdot (\vec{v} \times \vec{u}).$$

Substitute and take root:

$$\boxed{|\vec{w}| = \frac{\sqrt{(\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) - (1/c^2)(\vec{v} \times \vec{u}) \cdot (\vec{v} \times \vec{u})}}{1 - \vec{v} \cdot \vec{u}/c^2}}. \quad (4)$$

(d)

Set $\vec{v} = \vec{u}_1 = (c/2, 0, 0)$, $\vec{u} = \vec{u}_2 = (c/2)(1, 1, 0)/\sqrt{2}$. Compute building blocks:

$$\vec{v} \cdot \vec{u} = \frac{c^2}{4\sqrt{2}}.$$

$$(\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) = \frac{c^2}{2} \left(1 - \frac{1}{\sqrt{2}} \right).$$

$$|\vec{v} \times \vec{u}|^2 = v^2 u^2 - (\vec{v} \cdot \vec{u})^2 = \frac{c^4}{16} - \frac{c^4}{32} = \frac{c^4}{32}.$$

Insert into (4):

$$V_{\text{rel}}^2 = \frac{(c^2/2)(1 - 1/\sqrt{2}) - c^2/32}{(1 - 1/(4\sqrt{2}))^2}.$$

Numerical evaluation; $1/\sqrt{2} = 0.7071$, $1/(4\sqrt{2}) = 0.1768$:

$$\text{Numerator}/c^2 = 0.5 \times 0.2929 - 0.03125 = 0.11520.$$

$$\text{Denominator} = (0.8232)^2 = 0.67770.$$

$$V_{\text{rel}}^2/c^2 = 0.11520/0.67770 = 0.17000.$$

$$V_{\text{rel}}/c = 0.41231.$$

$$\Gamma = 1/\sqrt{0.83000} = 1.09764.$$

Doppler-equality (source S_2 , receiver S_1); $\hat{n}$ is light direction in S_1 :

$$\omega_{\text{obs}} = \frac{\omega_0}{\Gamma(1 - \vec{V}_{\text{rel}} \cdot \hat{n}/c)} = \omega_0.$$

Solve for $\cos \alpha = \hat{V}_{\text{rel}} \cdot \hat{n}$:

$$\cos \alpha = \frac{c}{V_{\text{rel}}} \left(1 - \frac{1}{\Gamma} \right).$$

Numerical:

$$1 - 1/\Gamma = 1 - 0.91105 = 0.08895.$$

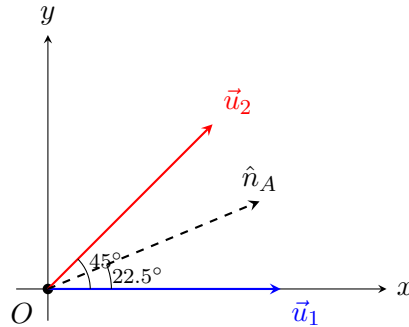
$$\cos \alpha = 0.08895/0.41231 = 0.21574.$$

$$\alpha = \arccos[(c/V_{\text{rel}})(1 - 1/\Gamma)] \approx 77.55^\circ.$$

Direction in frame A. The Doppler-equality condition is symmetric under exchange of source and receiver. In A both stars move at speed $c/2$, so the configuration is invariant under reflection about the bisector of $\hat{u}_1, \hat{u}_2$. The light ray must lie on this symmetry axis:

$$\hat{n}_A \propto \hat{u}_1 + \hat{u}_2 = \left(1 + \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right),$$

i.e. at angle $\arctan[(1/\sqrt{2})/(1 + 1/\sqrt{2})] = 22.5^\circ$ to the x -axis.



Question 4 — Constant force, 4-force, synchrotron

Problem. (a) For constant $\vec{F}$ from rest find $\vec{v}(t)$ and show $|\vec{v}| \rightarrow c$. (b) Show $F^\alpha = (\gamma \vec{F} \cdot \vec{v}/c, \gamma \vec{F})$ with $\vec{F} = m\gamma^3[\vec{a} + \vec{v} \times (\vec{v} \times \vec{a})/c^2]$. (c) Compare radiative loss in linear vs circular accelerators. (d) Choose particle/machine to minimise loss.

(a)

Integrate $\vec{F} = d(\gamma m \vec{v})/dt$ from rest:

$$\gamma m \vec{v} = \vec{F}t.$$

Square and use $\gamma^2 v^2 = v^2/(1 - v^2/c^2)$:

$$\frac{m^2 v^2}{1 - v^2/c^2} = F^2 t^2.$$

Clear the denominator:

$$m^2 v^2 = F^2 t^2 (1 - v^2/c^2).$$

Collect v^2 :

$$v^2 (m^2 + F^2 t^2 / c^2) = F^2 t^2.$$

Solve and align direction:

$$\vec{v}(t) = \frac{\vec{F}t/m}{\sqrt{1 + (Ft/(mc))^2}}.$$

Take $t \rightarrow \infty$: numerator $\sim Ft/m$, denominator $\sim Ft/(mc)$, ratio $\rightarrow c$:

$$\lim_{t \rightarrow \infty} |\vec{v}| = c.$$

(b)

Time component, $F^0 = dp^0/d\tau$ with $dm/dt = 0$:

$$F^0 = \gamma \frac{d(m\gamma c)}{dt} = \frac{\gamma}{c} \frac{dE}{dt}.$$

Apply work-energy $dE/dt = \vec{F} \cdot \vec{v}$:

$$F^0 = \gamma \vec{F} \cdot \vec{v} / c.$$

Spatial: $F^i = \gamma dp^i/dt = \gamma F^i$:

$$F^\alpha = (\gamma \vec{F} \cdot \vec{v} / c, \gamma \vec{F}).$$

Now $\vec{F} = d(m\gamma \vec{v})/dt$:

$$\vec{F} = m \left(\frac{d\gamma}{dt} \vec{v} + \gamma \vec{a} \right).$$

Use $d\gamma/dt = \gamma^3 \vec{v} \cdot \vec{a} / c^2$:

$$\vec{F} = m\gamma \vec{a} + \frac{m\gamma^3}{c^2} (\vec{v} \cdot \vec{a}) \vec{v}.$$

Apply BAC–CAB with $\vec{A} = \vec{B} = \vec{v}$, $\vec{C} = \vec{a}$:

$$\vec{v} \times (\vec{v} \times \vec{a}) = (\vec{v} \cdot \vec{a}) \vec{v} - v^2 \vec{a}.$$

Rearrange:

$$(\vec{v} \cdot \vec{a}) \vec{v} = \vec{v} \times (\vec{v} \times \vec{a}) + v^2 \vec{a}.$$

Substitute:

$$\vec{F} = m\gamma \vec{a} + \frac{m\gamma^3}{c^2} [\vec{v} \times (\vec{v} \times \vec{a}) + v^2 \vec{a}].$$

Group $\vec{a}$ terms; use $1 + \gamma^2 v^2/c^2 = \gamma^2$:

$$\vec{F} = m\gamma\left(1 + \frac{\gamma^2 v^2}{c^2}\right)\vec{a} + \frac{m\gamma^3}{c^2}\vec{v} \times (\vec{v} \times \vec{a}).$$

$$\boxed{\vec{F} = m\gamma^3\left[\vec{a} + \frac{\vec{v} \times (\vec{v} \times \vec{a})}{c^2}\right].}$$

(c)

Linear case ($\vec{a} \parallel \vec{v}$): $\vec{v} \times \vec{a} = 0$:

$$F = m\gamma^3 a.$$

Compute the invariant directly using the Minkowski-norm of F^α (for $\vec{F} \parallel \vec{v}$, $F^\alpha F_\alpha = F^2$):

$$\frac{dp^\alpha}{d\tau} \frac{dp_\alpha}{d\tau} = F^2 = m^2 \gamma^6 a^2.$$

Insert into Larmor:

$$P_{\text{lin}} = \frac{q^2}{6\pi\epsilon_0 m^2 c^3} m^2 \gamma^6 a^2 = \frac{q^2 \gamma^6 a^2}{6\pi\epsilon_0 c^3}.$$

Circular case ($\vec{a} \perp \vec{v}$): $\vec{v} \times (\vec{v} \times \vec{a}) = -v^2 \vec{a}$:

$$F = m\gamma^3(1 - v^2/c^2)a = m\gamma a.$$

Then $\vec{F} \cdot \vec{v} = 0$, $F^\alpha F_\alpha = \gamma^2 F^2$:

$$\frac{dp^\alpha}{d\tau} \frac{dp_\alpha}{d\tau} = \gamma^2 m^2 \gamma^2 a^2 = m^2 \gamma^4 a^2.$$

$$P_{\text{circ}} = \frac{q^2 \gamma^4 a^2}{6\pi\epsilon_0 c^3}.$$

Ratio at fixed (γ, a) :

$$\boxed{\frac{P_{\text{lin}}}{P_{\text{circ}}} = \gamma^2.}$$

(d)

Larmor scaling at fixed energy in a ring ($\gamma = E/mc^2$):

$$P_{\text{circ}} \propto \frac{q^2 \gamma^4 a^2}{m^2 c^3} \propto \frac{E^4}{m^6 c^8}.$$

Mass ratio: $m_\mu/m_e \approx 207$. Suppression in a ring: $(m_\mu/m_e)^6 \approx 7.9 \times 10^{13}$. In a linear machine the curvature term vanishes, removing the γ^4 factor entirely. Combine: choose

$$\boxed{\text{muon in a linear accelerator.}}$$

Reason: (i) linear geometry eliminates the synchrotron γ^4 enhancement (no centripetal acceleration); (ii) the muon's larger rest mass suppresses Larmor by $1/m^2$ at fixed force, equivalently $1/m^6$ at fixed energy.